# Mock 02 Solutions

## SECTION A

Answer **ALL** questions in this section.

1. (a) State precisely the definition of the divisibility relation $a\,|b$ on the integers and prove that the greatest common divisor, $d$, of any pair of integers, $a, b$, can be expressed as a linear combination of $a$ and $b$, i.e.

$$\exists m, n \in \mathbb{Z} \quad d = ma + nb.$$

*Hint: Consider the set, $S$, of all postive linear combinations of $a$ and $b$, i.e.*

$$S = \{xa + yb \,|\, x, y \in \mathbb{Z} \ \& \ xa + yb > 0\}.$$

(6 marks)

**Solution:** We say $a$ divides $b$, and write $a|b$, if and only if there exists an integer $c$ such that $b = ac$. Consider

$$S = \{xa + yb \,|\, x, y \in \mathbb{Z} \ \& \ xa + yb > 0\}.$$

By the well-ordered principle $S$ contains a least element. Call this $e$, and we have $e = ua + vb$, for some $u, v \in \mathbb{Z}$. We can show that $e$ is a common divisor of $a$ and $b$. Suppose $e$ does not divide $a$, then we can divide $a$ as

$$a = qe + r,$$

where the remainder $r$ satisfies $0 < r < e$. Then we can say

$$r = a - qe = a - q(ua + vb) = (1 - qu)a + vb.$$

This shows that $r \in S$, since it is a positive linear combination of $a$ and $b$. But $r < e$. This contradicts the definition of $e$ as the least elenment of $S$. Therefore $e$ divides $a$. Using a similar argument we can show that $e$ divides $b$.

So $e$ is a common divisor of $a$ and $b$, and since $e = ua + vb$, every common divisor of $a$ and $b$ divides $e$. Therefore $e$ is the greatest common divisor of $a$ and $b$.

(b) Use the principle of mathematical induction to prove that

$$\forall n \geq 1 \quad 13 \,|\, \left(5^{4n} - 1\right).$$

You should point out in your argument where you make use of the linear combinations result from part (a) above. (5 marks)

**Solution:** When $n = 1$, $5^4 - 1 = 624 = 48 \cdot 13$, and clearly $13|624$. So the base case is true. Let us assume that $13|5^{4k} - 1$, for some $k \geq 1$, and then note that

$$\begin{aligned} 5^{4(k+1)} - 1 &= 625 \cdot 5^{4k} - 1 \\ &= 625 \cdot (2^{4k} - 1) + 624. \end{aligned}$$

Note that this last expression is a linear combination of integers divisible by 13, so we have $13|(5^{4(k+1)} - 1)$. Hence, by induction, $13 \mid (5^{4n} - 1)$ for all $n \geq 1$.

(c) Prove that there are infinitely many prime numbers. State clearly any results about divisibility that you rely on. (10 marks)

**Solution:** We present Euclid's proof by contradiction. Assume, on the contrary, that there are only a finite number of primes, and that they can be listed as $p_1, p_2, \ldots, p_n$.

Then consider the integer $N$ defined by

$$N = \left( \prod_{i=1}^{n} p_i \right) + 1.$$

By the Fundamental theorem of arithmetic $N$ factorizes uniquely into primes, so we can say that there is some $j$, with $1 \leq j \leq n$ and $p_j|N$.

Rewriting the above equation as

$$1 = N - \left( \prod_{i=1}^{n} p_i \right),$$

we see that $p_j|1$ as $1$ is a linear combination of integers, both divisible by $p_j$.

But $p_j|1$ contradicts the fact that $p_j > 1$ as $p_j$ is prime.

So we conclude that there are infinitely many primes.

(d) Prove that if an integer of the form $2^n - 1$ is prime then it must be the case that the exponent $n$ is also prime. (4 marks)

**Solution:** We prove the contrapositive.

Assume that $n$ is not a prime, i.e. that $n = rs$ for integers $r$ and $s$, both strictly greater than 1.

Then we can factorize $2^n - 1$ as

$$\begin{aligned} 2^n - 1 &= 2^{rs} - 1 \\ &= (2^r)^s - (1)^s \\ &= (2^r - 1) \sum_{j=0}^{s-1} \left(2^{rj}\right) \end{aligned}$$

The conditions on $r, s$ imply that this is a genuine factorization, i.e. $1 < 2^r - 1 < 2^n + 1$, and hence $2^n - 1$ is not prime.

## SECTION A

2. (a) Let $G$ be a non-empty set and $*$ a binary operation on $G$, i.e.

$$\forall\, g_1, g_2 \in G \quad g_1 * g_2 \in G.$$

State the three extra conditions that the pair $(G, *)$ needs to satisfy in order to be called a group and explain their meaning. Illustrate each condition with an example drawn from the group $(\mathsf{GL}(2, \mathbb{R}), \times)$. (8 marks)

**Solution:** The operation is associative on $G$, i.e. for all $g, h, k \in G$,

$$g * (h * k) = (g * h) * k.$$

This means that no brackets are needed in products of group elements. Multiplication of matrices, the operation in GL$(2, \mathbb{R})$, is known to be associative, e.g. for all such matrices $A, B, C$ we have $A(BC) = (AB)C$.

There exists an identity element $e \in G$ satisfying for all $g \in G$, $e * g = g * e = g$. When multiplied by any element the identity leaves the other element unchanged. In GL$(n, \mathbb{R})$ the identity is $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ as for all $2 \times 2$ matrices $A$, we have $IA = AI = A$.

For all elements $g \in G$ there exists an inverse element $g^{-1} \in G$ satisfying $g * g^{-1} = g^{-1} * g = e$, the identity element of $G$. In , GL$(n, \mathbb{R})$, the inverse elements are the *inverse* matrices in the linear algebra sense. The inverse of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is the matrix $\dfrac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

(b) The external direct product, $\mathbb{Z}_2 \times \mathbb{Z}_2$, is the group consisting of the four ordered pairs, $\{(0, 0), (0, 1), (1, 0), (1, 1)\}$, of elements of $\mathbb{Z}_2$, under the usual operation in an external direct product.

(i) Clearly define the operation in an external direct product and write down the Cayley table for the group $\mathbb{Z}_2 \times \mathbb{Z}_2$. Also write down the Cayley table for the group $\mathbb{Z}_4$, the integers under addition modulo 4. (3 marks)

(ii) From the two Cayley tables point out one feature that shows these two groups have a different structure. (2 marks)

**Solution:**

(i) Given two groups $(G, *)$ and $(H, \circ)$, their external direct product, $G \times H$, is the group formed by the ordered pairs

$$G \times H = \{(g, h) \,:\, g \in G, h \in H\},$$

under the operation defined by

$$(g_1, h_1)(g_2, h_2) = (g_1 * g_2, h_1 \circ h_2).$$

The Cayley tables are

| $\mathbb{Z}_2 \times \mathbb{Z}_2$ | $(0,0)$ | $(0,1)$ | $(1,0)$ | $(1,1)$ |
|---|---|---|---|---|
| $(0,0)$ | $(0,0)$ | $(0,1)$ | $(1,0)$ | $(1,1)$ |
| $(0,1)$ | $(0,1)$ | $(0,0)$ | $(1,1)$ | $(1,0)$ |
| $(1,0)$ | $(1,0)$ | $(1,1)$ | $(0,0)$ | $(0,1)$ |
| $(1,1)$ | $(1,1)$ | $(1,0)$ | $(0,1)$ | $(0,0)$ |

and

| $\mathbb{Z}_4$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0 | 0 | 1 | 2 | 3 |
| 1 | 1 | 2 | 3 | 0 . |
| 2 | 2 | 3 | 0 | 1 |
| 3 | 3 | 0 | 1 | 2 |

(ii) In $\mathbb{Z}_2 \times \mathbb{Z}_2$ every element $x$ satisfies $x + x = e$, i.e. has order 1 or 2. In $\mathbb{Z}_4$ this is not so.

(c) Which matrices are elements of the group GL$(n, \mathbb{Q})$? Prove that this is a group under the operation of matrix multiplication. You can make use of any facts about matrices from linear algebra, provided you clearly state these in your answer. *Note: the field of coefficients is $\mathbb{Q}$, not $\mathbb{R}$.* (7 marks)

**Solution:** GL$(n, \mathbb{Q})$ is the group of $n \times n$ matrices with rational coefficients and non-zero determinant under the operation of multiplication. The determinant satisfies

$$\det(AB) = \det(A)\det(B).$$

So if $\det(A)$ and $\det(B)$ are non-zero then so is $\det(AB)$.

Secondly, since matrix multiplication just consists of repeated multiplications and additions involving the entries of the two matrices, the entries of the product matrix will again all be rational numbers.

Therefore GL$(n, \mathbb{Q})$ is closed under matrix multiplication. Matrix multiplication is known to be associative, i.e. for all $n \times n$ matrices $A, B, C$,

$$A(BC) = (AB)C.$$

The identity element is the similarly named $n \times n$ *identity* matrix

$$I = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix},$$

which satisfies for all $n \times n$ matrices $A$, $AI = IA = A$. The entries of $I$ are clearly rational, and it has determinant $1$. Hence $I \in$ GL$(n, \mathbb{Q})$.

Lastly, every $n \times n$ matrix $A$ with non-zero determinant has an inverse (w.r.t. mat. mult.) which has determinant $1/\det(A)$, which is also non-zero. The co-factor formula that gives the inverse again just involves many sums and products of the entries, and so also preserves the rational nature of the entries. Hence $A^{-1}$ is an element of GL$(n, \mathbb{Q})$.

(d) Consider the set of $2 \times 2$ matrices $H \subset$ GL$(2, \mathbb{Q})$ given by

$$H = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \;:\; a, b, c \in \mathbb{Q} \;\&\; ac \neq 0 \right\}.$$

Prove that $H$ forms a subgroup of GL$(2, \mathbb{Q})$. (5 marks)

**Solution:** We will show that the conditions of the subgroup theorem are satisfied by $H$.

Note that the matrix in $H$ shown in the definition has determinant equal to $ac \neq 0$ so $H$ is indeed a subset of GL$(2, \mathbb{Q})$. The identity matrix is clearly an element of $H$ (use $a = c = 1, b = 0$). So it just remains to prove that $H$ is closed under matrix multiplication and contains the necessary inverse matrices. Checking closure we find

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \begin{pmatrix} d & e \\ 0 & f \end{pmatrix} = \begin{pmatrix} ad & ae + bf \\ 0 & cf \end{pmatrix} \in H.$$

Checking for inverses we find

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}^{-1} = \frac{1}{ac} \begin{pmatrix} c & -b \\ 0 & a \end{pmatrix} \in H.$$

**END OF SECTION A**

## SECTION B

Answer **TWO** questions in this section.

3. (a) State the definition of the congruence relation $a \equiv b \pmod{n}$. How does it relate to the smallest positive remainders left by $a$ and $b$ upon division by $n$? (3 marks)

> **Solution:** The definition of $a \equiv b \pmod{n}$ is that $n|(a-b)$.
>
> And $a \equiv b \pmod{n}$ is equivalent to $a$ and $b$ leaving the same small positive remainder, $r$ with $0 \le r < n$, upon division by $n$.

(b) Let $a, n \in \mathbb{Z}$ with $n > 0$. Prove that a multiplicative inverse, $a^{-1}$, for $a$ modulo $n$ exists if and only if $\gcd(a, n) = 1$. Find the inverse, $y$, for $13$ modulo 100, that lies in the range $1 \le y \le 99$. (10 marks)

> **Solution:** We establish a chain of equivalences linking the two conditions.
>
> $$\begin{aligned} \gcd(a,n) = 1 &\Leftrightarrow \exists r, s \in \mathbb{Z}\ ar + ns = 1, \text{ by the Euclidean alogirhtm and Bezout's identity} \\ &\Leftrightarrow \exists r, s \in \mathbb{Z}\ ar - 1 = n(-s) \\ &\Leftrightarrow \exists r \in \mathbb{Z}\ n|ar - 1, \text{ by the defn. of divisibility} \\ &\Leftrightarrow \exists r \in \mathbb{Z}\ ar \equiv 1 \pmod{n}, \text{by the defn. of congruence} \\ &\Leftrightarrow a \text{ is invertible modulo } n \end{aligned}$$
>
> To find $13^{-1}$ modulo $100$ we run the extended Euclidean algorithm on the pair $13, 100$ to find:
>
> $$\begin{aligned} 100 &= 7 \cdot 13 + 9 \\ 13 &= 1 \cdot 9 + 4 \\ 9 &= 2 \cdot 4 + 1 \end{aligned}$$
>
> then back subsitution provides:
>
> $$\begin{aligned} 1 &= 9 - 2 \cdot 4 \\ &= 9 - 2 \cdot (13 - 9) \\ &= -2 \cdot 13 + 3 \cdot 9 \\ &= -2 \cdot 13 + 3 \cdot (100 - 7 \cdot 13) \\ &= 3 \cdot 100 - 23 \cdot 13. \end{aligned}$$
>
> This shows that $-23$ is an inverse for $13$ modulo $100$. The inverse in the required range is thus $y = 77$, as $77 \equiv -23 \pmod{100}$.

(c) State the definition of the Euler totient function $\phi$ and prove that for any prime $p$ and positive integer $n$, that $\phi$ satisfies

$$\phi(p^n) = p^{n-1}(p-1).$$

(7 marks)

**Solution:** The definition of $\phi(m)$ is the number of integers $j$ such that $1 \leq j \leq m$ and $\gcd(j, m) = 1$.

Of the integers $j$ in the range $1 \leq j \leq p^n$, the ones that are not coprime to $p^n$ are those integers $j$ such that $p|j$. This is because $p$ is a prime so the only divisors of $p^n$ are other powers of $p$.

Within the sequence $1, 2, 3 \ldots, p^n$ the multiples of $p$ are

$$p, 2p, 3p, \ldots, p^{n-1}p = p^n,$$

so there are $p^{n-1}$ of these, and so the number of $j$ that **are** coprime to $p^n$ is

$$\begin{aligned}\phi(p^n) &= p^n - p^{n-1} \\ &= p^{n-1}(p-1),\end{aligned}$$

as required.

(d) Write down the multiplicative formula for $\phi(n)$ that gives its value based on the prime factorization of $n$, where $n \in \mathbb{Z}^+$. Use the formula to calculate $\phi(100)$. (5 marks)

**Solution:** The full formula for $\phi$ is

$$\phi(n) = \prod_{i=1}^{r} \left(p_i^{a_i - 1}(p_i - 1)\right),$$

where $n$ has the prime factorization $n = \prod_{i=1}^{r} p_i^{a_i}$, for distinct primes $p_i$ and non-zero exponents $a_i$.

We calculate $\phi(100)$ using this and the prime factorization of $100$.

$$\begin{aligned}\phi(100) &= \phi(2^2 \cdot 5^2) \\ &= 2^1 \cdot (2-1) \cdot 5^1 \cdot (5-1) \\ &= 2 \cdot 5 \cdot 4 \\ &= 40\end{aligned}$$

## SECTION B

4. (a) State Lagrange's theorem on the orders of subgroups of a finite group $G$. (3 marks)

**Solution:** Anything equivalent to: If $G$ is a finite group and $H$ a subgroup of $G$ then $|H|$ divides $|G|$.

(b) Let $H$ be a subgroup of a finite group $G$.

(i) State the definition of the left and right cosets of $H$ in $G$. (2 marks)

(ii) Let $g_1, g_2 \in G$. Prove that the left-cosets $g_1H$ and $g_2H$ are either equal or disjoint, i.e.

$$g_1H = g_2H \quad \text{or} \quad g_1H \cap g_2H = \emptyset.$$

(6 marks)

(iii) Prove that all cosets of $H$ in $G$ contain the same number of elements. (3 marks)

(iv) Then show how parts (ii) and (iii) above can be used to prove Lagrange's theorem. (4 marks)

**Solution:**

(i) The left-coset of $H$ in $G$ with representative $g \in G$ is the subset $gH$ of $G$ defined by

$$gH = \{gh \,:\, h \in H\}.$$

The right-coset of $H$ in $G$ with representative $g \in G$ is the subset $Hg$ of $G$ defined by

$$Hg = \{hg \,:\, h \in H\}.$$

(ii) Suppose that $x \in g_1H \cap g_2H$. Therefore there exists $h_1, h_2 \in H$ such that $x = g_1h_1 = g_2h_2$ which implies that $g_1 = g_2h_2h_1^{-1}$ and $g_2 = g_1h_1h_2^{-1}$.
Then for all $h \in H$ we have

$$g_1h = g_2h_2h_1^{-1}h \in g_2H,$$

as the product $h_2h_1^{-1}h$ is clearly in $H$ as $H$ is a subgroup. So $g_1H \subseteq g_2H$.
Similarly, for all $h \in H$ we have

$$g_2h = g_1h_1h_2^{-1}h \in g_1H.$$

So $g_2H \subseteq g_1H$. Therefore $g_1H = g_2H$. This proves that if the intersection of $g_1H$ and $g_2H$ is non-empty (there is such an $x$ as above) then the two cosets are equal.

So in conclusion any two cosets are either disjoint or equal.

(iii) We will show that all cosets of $H$ contain $|H|$ elements by exhibiting the bijection $\phi : H \to gH$ defined by $\phi(h) = gh$.

That this is surjective is clear from the definition of $gH$.

Suppose that $\phi(h_1) = \phi(h_2)$, i.e. $gh_1 = gh_2$. Applying the group cancellation law (multiplying on left by $g^{-1}$) shows that $h_1 = h_2$. So $\phi$ is also injective, and hence a bijection. Therefore all cosets contain the same number, $|H|$, of elements.

(iv) The collection of cosets of $H$ in $G$ forms a partition of $G$, meaning every element of $G$ is contained in some coset, for if $g \in G$ then $g \in gH$ as $g = ge \in gH$ as $e \in H$. Also different subsets of the partition are disjoint (from part (ii)).

So the total number of elements of $G$ can be found by summing the number of elements in each distinct coset. Part (iii) shows that every coset has the same number of elements, namely $|H|$.

So we get the result of Lagrange's theorem that

$$|G| = [G : H]|H|,$$

where $[G : H]$, the index of $H$ in $G$ is the number of distinct cosets of $H$ in $G$.

(c) The dihedral group $D_8$ is generated by the pair of elements $r, s$ which are subject to the relations $r^8 = e$, $s^2 = e$ and $sr = r^{-1}s$. Consider the subgroup $H$ of $D_8$ given by

$$H = \left\{e, r^2, r^4, r^6\right\}.$$

(i) Work out the elements of each left coset of $H$ in $D_8$. (4 marks)

(ii) Give an example of a subgroup $K$ of $D_8$ and an element $x \in D_8$ for which

$$xK \neq Kx.$$

(3 marks)

**Solution:**

(i) The distinct cosets are (one on each line)

$$
\begin{aligned}
H = r^2H = r^4H = r^6H &= \{e, r^2, r^4, r^6\} \\
rH = r^3H = r^5H = r^7H &= \{r, r^3, r^5, r^7\} \\
sH = r^2sH = r^4sH = r^6sH &= \{s, r^2s, r^4s, r^6s\} \\
rsH = r^3sH = r^5sH = r^7sH &= \{rs, r^3s, r^5s, r^7s\}.
\end{aligned}
$$

(ii) An example is $K = \{e, s\}$ and $x = r$. For then we have

$$rK = \{r, rs\} \text{ and } Kr = \{r, r^7s\},$$

which are different since $rs \neq r^7s$.

## SECTION B

5. (a) Consider the congruence

$$21x \equiv 14 \pmod{49}.$$

User relevant result(s) from the theory of congruences to find all the solutions. (9 marks)

**Solution:** Using the result about linear congruences we see that $\gcd(21, 49) = 7$ and $7|14$ so there are seven solutions to the congruence given by

$$t + 7i, \quad (i = 0, 1, 2, 3, 4, 5, 6)$$

where $t$ is the unique solution $0 \leq t \leq 6$ to the reduced congruence

$$3x \equiv 2 \pmod{7}.$$

From the Euclidean algorithm (or simple observation) we see that $3^{-1} \equiv 5 \pmod{7}$ so that

$$t \equiv 5 \times 2 \equiv 10 \equiv 3 \pmod{7}.$$

So the seven solutions to the original congruence are given by

$$x \equiv 3, 10, 17, 24, 31, 38, 45 \pmod{49}.$$

(b) Use the Chinese Remainder Theorem to describe the integers $x$ that satisfy all three of the following congruences simultaneously,

$$\begin{aligned} x &\equiv 4 \pmod{7} \\ x &\equiv 3 \pmod{13} \\ x &\equiv 9 \pmod{17}. \end{aligned}$$

Your final answer should be in the form of a single congruence class for $x$ modulo an appropriate modulus. (10 marks)

**Solution:** We use the Chinese Remainder theorem from the handout. Firstly the three moduli $m_1 = 7,\ m_2 = 13,\ m_3 = 17$ are distinct primes, hence pairwise coprime so the theorem applies.

Let $M = 7 \times 13 \times 17 = 1547$. Then we have

$$M_1 = 221,\ M_2 = 119,\ M_3 = 91,$$

and the multiplicative inverses $M_i'$ of $M_i$ modulo $m_i$ are obtained from the following:

$$\begin{aligned} M_1' &\equiv (221)^{-1} \pmod 7 \\ &\equiv 4^{-1} \pmod 7 \\ &\equiv 2, \pmod 7 \end{aligned}$$

and then

$$\begin{aligned} M_2' &\equiv (119)^{-1} \pmod{13} \\ &\equiv 2^{-1} \pmod{13} \\ &\equiv 7, \pmod{13} \end{aligned}$$

and then

$$\begin{aligned} M_3' &\equiv (91)^{-1} \pmod{17} \\ &\equiv 6^{-1} \pmod{17} \\ &\equiv 3, \pmod{19} \end{aligned}$$

The solutions $x$ to the system of congruences in the question are all of the integers in the congruence class

$$\begin{aligned} x &\equiv 4M_1M_1' + 3M_2M_2' + 9M_3M_3' \pmod{1547} \\ &\equiv 4 \cdot 221 \cdot 2 + 3 \cdot 119 \cdot 7 + 9 \cdot 91 \cdot 3 \pmod{1547} \\ &\equiv 6724 \equiv 536 \pmod{1547}. \end{aligned}$$

(c) Use the Legendre symbol, the law of quadratic reciprocity and other relevant properties to show that there are integer solutions to the congruence

$$x^2 \equiv 307 \pmod{601}.$$

(You can use the fact that $307$ and $601$ are both prime.) (6 marks)

**Solution:** Solutions $x$ exist if and only if $307$ is a quadratic residue modulo $601$, i.e. iff the Legendre symbol satisfies $(307|601) = +1$.

We evaluate the Legendre symbol as follows:

$$\begin{aligned} (307|601) &= (601|307)\,, &&\text{quadratic reciprocity, } 601 \equiv 1 \pmod 4 \\ &= (294|307)\,, &&\text{since } 601 \equiv 294 \pmod{307} \end{aligned}$$

$$
\begin{aligned}
&= \left(2 \cdot 3 \cdot 7^2 | 307\right), \quad \text{factorizing } 294 \\
&= (2|307) \cdot (3|307) \cdot (7|307)^2 \quad \text{by the mult. prop. of the Legendre symbol} \\
&= (2|307) \cdot (3|307), \ \text{since any Legendre symbol is } \pm 1 \\
&= (307|3), \text{quad. recip., } 307 \equiv 3 \pmod 4, (2|307) = -1 \text{ since } 307 \equiv 3 \pmod 8 \\
&= (1|3), \ \text{since } 307 \equiv 1 \pmod 3 \\
&= 1, \ \text{as } 1 \text{ is always a quad. residue}
\end{aligned}
$$

Thus, $(307|601) = +1$, and therefore $307$ is a quadratic residue modulo $601$, and so solutions $x$ exists to the congruence in question.

## SECTION B

6. (a) Give the definition of a **normal subgroup**. (3 marks)

> **Solution:** A subgroup $H$ of a group $G$ is normal in $G$ iff for all $g \in G$ we have $gH = Hg$, i.e. left and right cosets of $H$ in $G$, represented by the same element $g$, are equal.

(b) Prove that a subgroup $N$ of a group $G$ is normal in $G$ if and only if

$$\forall g \in G \quad gNg^{-1} \subset N.$$

(5 marks)

> **Solution:** Suppose $N$ is normal in $G$. Then we have
>
> $$\begin{aligned} &\forall g \in G\ gN = Ng \\ \Rightarrow &\forall g \in G\, \forall n \in N\, \exists m \in N\ gn = mg \\ \Rightarrow &\forall g \in G\, \forall n \in N\, \exists m \in N\ gng^{-1} = m \\ \Rightarrow &\forall g \in G\, \forall n \in N\, gng^{-1} \in N \\ \Rightarrow &\forall g \in G\ gNg^{-1} \subset N \end{aligned}$$
>
> Conversely, suppose that $\forall g \in G\ gNg^{-1} \subset N$. This is equivalent to
>
> $$\forall g \in G\, \forall n \in N\, \exists m \in N\ gng^{-1} = m. \qquad (*)$$
>
> Consider any $gn \in gN$. From $(*)$ we know $gng^{-1} = m \in N$ for some $m \in N$, hence $gn = mg \in Ng$. Therefore $gN \subset Ng$. Now consider any $ng \in Ng$. From $(*)$ (but applied to $g^{-1}$ rather than $g$) we know that $g^{-1}ng = m \in N$ for some $m \in N$, hence $ng = gm \in gN$. Therefore $Ng \subset gN$, and so $gN = Ng$, as required.

(c) If $G$ is a group and $g \in G$ then the centralizer, $C(g)$, of $g$ in $G$ is the subgroup of $G$ defined by

$$C(g) = \{h \in G \mid hg = gh\},$$

i.e. all the elements from $G$ that commute with $g$.

Prove that if $g$ generates a normal subgroup of $G$, i.e. $\langle g \rangle$ is normal in $G$, then the centralizer $C(g)$ is also normal in $G$. (9 marks)

**Solution:** Let $g \in G$. Suppose that $\langle g \rangle$ is normal in $G$, so by part (c) we have $\forall h \in G\, h\langle g \rangle h^{-1} \subset \langle g \rangle$. This means that

$$\forall h \in G\, \forall n \in \mathbb{Z}\, \exists m \in \mathbb{Z}\ hg^n h^{-1} = g^m. \qquad (*)$$

To prove that $C(g)$ is normal in $G$ we need to show that for all $x \in G$ we have $xC(g)x^{-1} \subset C(g)$. So let $x \in G$ and let $h \in C(g)$. We now need to show that $xhx^{-1}$ is in $C(g)$, i.e. that it commutes with $g$. This is proved by the following:

$$\begin{aligned} xhx^{-1}g &= xhg^m x^{-1}, \text{ since } x^{-1}g = g^m x^{-1}, \text{ for some } m \in \mathbb{Z}, \text{ by } (*) \\ &= xg^m hx^{-1}, \text{ since } h \text{ commutes with } g \text{ as } h \in C(g) \\ &= gxhx^{-1}. \text{ as } x^{-1}g = g^m x^{-1} \Leftrightarrow gx = xg^m \end{aligned}$$

This chain of equations shows that $xhx^{-1}$ commutes with $g$, as required. Therefore $C(g)$ is indeed normal in $G$.

(d) Let $x \in \mathbb{R}$ with $x \neq 0, 1$ and let $X$ be the matrix

$$X = \begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix},$$

considered as an element of the group GL$(2, \mathbb{R})$. Identify exactly which matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ belong to the centralizer $C(X)$. Your conclusion should be given in terms of conditions on the entries $a, b, c, d$.

(8 marks)

**Solution:** A matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is in $C(X)$ iff it commutes with $X$. Examining this condition we find:

$$\begin{aligned} &\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ \Leftrightarrow &\begin{pmatrix} ax & b \\ cx & d \end{pmatrix} = \begin{pmatrix} ax & bx \\ c & d \end{pmatrix} \\ \Leftrightarrow\ & b = bx \text{ and } c = cx \\ \Leftrightarrow\ & b = 0 \text{ and } c = 0, \text{ since } x \neq 1 \end{aligned}$$

From this we see that

$$C(X) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}(2, \mathbb{R})\ :\ b = c = 0 \right\},$$

i.e. the commutator of $X$ consists of all the diagonal matrices.

**END OF SECTION B**
**END OF QUESTIONS**